Analytical Activity

1. Gradient Magnitude and Gradient Direction

Given,

$G_x = 24$, $G_y = 18$

Gradient Magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

$$G = \sqrt{24^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30$$

$G = 30$

Gradient Direction

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right) = \tan^{-1}\left(\frac{18}{24}\right) =$$

$\theta = 36.87^\circ$

2. Non-Maximum Suppression

Given

Pixel	P_1	P_2	P_3	P_4	P_5
magnitude	25	50	90	60	30

Gradient direction is horizontal, so each pixel is compared with its left and right neighbours.

Process

- $P_1 = 25$: Boundary pixel → suppress → 0
- $P_2 = 50$: compare 25 and 90 → 50 is not maximum → 0
- $P_3 = 90$: compare 50 and 60 → 90 is maximum → 90
- $P_4 = 60$: compare 90 and 30 → 60 is not maximum → 0
- $P_5 = 30$: Boundary pixel → suppress → 0

Final Result

0, 0, 90, 0, 0

3. Double Thresholding

Given:

- High threshold = 100
- low threshold = 50

Pixel	magnitude	classification
P_1	135	strong edge
P_2	90	weak edge
P_3	45	non-edge
P_4	120	strong edge
P_5	70	weak edge
P_6	20	non-edge

Therefore:

Strong: P_1, P_4 — if magnitude ≥ 100

weak: P_2, P_5 — if magnitude 50-99

Non-edge: P_3, P_6 — < 50 (magnitude)

4. Edge Tracking by Hysteresis

Given:

- High threshold = 100
- low threshold = 50

Step-1 - classification

$P_1 = 130 \rightarrow$ strong

$P_2 = 95 \rightarrow$ weak

$P_3 = 40 \rightarrow$ non-edge

$P_4 = 115 \rightarrow$ strong

$P_5 = 65 \rightarrow$ weak

$P_6 = 25 \rightarrow$ non-edge

$P_7 = 105 \rightarrow$ strong

Strong: P_1, P_4, P_7

weak: P_2, P_5

non-edge: P_3, P_6

Step-2: Hysteresis

- P_2 is connected to strong edge $P_1 \rightarrow$ keep P_2
- P_5 is not connected to any strong edge $\rightarrow$ Remove P_5

Final edge pixels: P_1, P_2, P_4, P_7

Hough circle Transform

1. Given:

$P_1 = (5,4)$ $\qquad$ $P(x,y)$, $C(a,b)$

$P_2 = (7,6)$

$C_1 = (2,4)$ $\qquad$ $d = \sqrt{(x-a)^2+(y-b)^2}$

$C_2 = (4,3)$

For center

$C_1(2,4)$

For $P_1(5,4)$ $\quad d = \sqrt{(5-2)^2+(4-4)^2} = \sqrt{9} = 3$

For $P_2 = (7,6)$ $\quad d = \sqrt{(7-2)^2+(6-4)^2} = \sqrt{25+4} = \sqrt{29} \approx 5.39$

For center

$C_2(4,3)$

For $P_1(5,4)$ $\quad d = \sqrt{(5-4)^2+(4-3)^2} = \sqrt{1^2+1^2} = \sqrt{2} \approx 1.41$

For $P_2(7,6)$ $\quad d = \sqrt{(7-4)^2+(6-3)^2} = \sqrt{9+9} = \sqrt{18} \approx 4.24$

center	P_1	P_2	matching radius
C_1	3	5.39	no
C_2	1.41	4.24	no

2. Radius

Given:

$P = (8,6)$, $C = (4,3)$

$$r = \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$r = 5$

3. Radius

Given:

$C = (5,4)$; $P = (8,8)$

$$r = \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16} = \sqrt{25} = 5$$

$r = 5$